



Modeling and Optimization Method of Laser Cladding Based on GA-ACO-RFR and GNSGA-II

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Abstract

Laser cladding is an environmentally friendly and reliable surface modification technology. The quality characteristics of the coating are directly affected by the process parameters of laser cladding. The reasonable selection of process parameters is essential to obtain high-quality coating. In this study, the single-track 15-5PH alloy coating was fabricated on the surface of 12Cr13 stainless steel. In view of the hybrid Genetic Algorithm and Ant Colony Optimization (GA-ACO) can effectively improve the prediction ability and robustness of Random Forest Regression (RFR), a prediction method of cladding layer quality characteristics based on GA-ACO-RFR was proposed. The fast non-dominated ranking genetic algorithm with elite strategy by introducing the Gaussian distribution crossover operator (GNSGA-II) was used to optimize the process parameters of laser cladding. The results showed that the multi-objective optimization method of laser cladding process parameters proposed in this paper can obtain high-quality laser cladding coating. This work demonstrated the potential of the proposed method in laser cladding process prediction and optimization.

Keywords Laser cladding · Coating · GA-ACO-RFR · GNSGA II · Modeling and optimization

1 Introduction

Surface modification can prepare strengthened coating on the surface of weaker and less expensive substrate materials through technical means, which can make the surface properties of the coating meet the higher requirements of complex working conditions. Due to its energy-saving, material saving and superior coating performance, it is widely used to enhance the service life of several engineering components or parts by improving their hardness, wear resistance, corrosion resistance or other mechanical properties [1]. With the rising of advanced manufacturing technology, the shortcomings of traditional surface modification technology are revealed, such as the bonding strength problem of thermal spraying technology [2], the environmental pollution problem of plating technology [3], and the efficiency problem of

thermochemical treatment technology [4], which are difficult to be solved. As a new surface modification technology, laser cladding technology can obtain a reinforced coating with a very low dilution rate on the surface of the substrate material and form metallurgical bonding with the substrate material, which is widely used in the fields of aerospace, mining machinery, machinery manufacturing and maintenance with the characteristics of low pollution emission, small deformation of the processed parts [5, 6].

Laser cladding technology can clad the same or dissimilar coating materials on the surface of the substrate according to the engineering requirements, both of which can form a metallurgical bond with the substrate and improve the performance of wear resistance, corrosion resistance, and high-temperature resistance of the substrate [6, 7]. However, the division of laser cladding material system and forming effect, the theoretical system of coupling relationship between light, powder, gas and substrate, the theoretical model of process parameters on coating forming effect, the acquisition of optimal process parameters under specific requirements affect the acquisition of stable coating forming quality and the development of laser cladding technology [8]. Therefore, to obtain coatings with good forming quality, it is crucial to establish prediction models and

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multi-objective optimization models of process parameters and coating geometric quality characteristics and mechanical properties.

To obtain good forming effect of laser cladding coating and complex mapping relationship between process parameters and coating quality characteristics, some scholars obtained coating quality characteristics data by designing orthogonal experiments, single-factor experiments, response surface experiments, etc., and obtained the relationship between process parameters and quality characteristics by ANOVA, signal-to-noise analysis, data fitting, etc., [9–11]. With the development of intelligent algorithms, some scholars try to establish the nonlinear relationship between process parameters and coating quality characteristics using algorithms such as neural networks, support vector machines and random forests [12, 13], and find that the accuracy of the model obtained by machine learning is higher than that of the traditional regression model through comparative analysis of prediction effects [14, 15]. The main reason is that because the relationship between process parameters and coating quality characteristics is nonlinear, empirical equations and response surfaces can achieve prediction and optimization under certain experimental volume conditions, but the accuracy of the results is limited. However, the proposed method is able to maintain good prediction performance despite the limited training data for machine learning [13].

Since algorithms such as neural networks, random forests, and support vector machines have hyperparameters, selecting a suitable optimization algorithm to optimize the settings of hyperparameters can effectively improve the accuracy and generalization ability of the algorithm. In this study, a Random Forest Regression prediction algorithm (RFR) based on Genetic Algorithm (GA) and Ant Colony Optimization (ACO) is proposed to establish the regression prediction modeling of laser cladding process parameters and coating quality characteristics. GA has better convergence, but poor local search ability, and is prone to a large number of redundant iterations due to the inability to use feedback information in the system [16, 17]; while ACO has good adaptability and strong local search ability, but low solution efficiency at the beginning of the search due to the lack of pheromones [18]. The genetic algorithm is used to search for the optimal solution to initialize the pheromone distribution of the ACO, and the randomly generated population of the genetic algorithm accelerates the convergence speed of the ACO and avoids falling into the local optimum.

More scholars choose to establish the regression model between process parameters and coating geometric quality characteristics [19, 20], and fewer scholars choose the mechanical properties of coatings [21, 22], mainly because the acquisition of coating mechanical properties requires high precision measurement equipment. Therefore, this

paper selects microhardness, aspect ratio and dilution rate of coatings, which are of great interest in engineering, as the evaluation indexes of coatings. Because the microhardness determines whether the coating can meet the requirements of working conditions, the aspect ratio of laser melting single-layer coating determines the surface flatness of multi-layer or multi-layer coating, and the dilution rate of the coating determines the bond strength of the coating and the substrate.

For process parameter optimization, some scholars use the traditional numerical analysis method to establish the comprehensive evaluation degree of coating quality characteristics to achieve the optimization of process parameters. The numerical analysis method can only select the best process parameters from the obtained data, but due to the limitation of the number of experiments, the process parameters found have a certain gap with the actual best process parameters. Therefore, some scholars try to use intelligent optimization algorithms for process parameter optimization, such as PSO [23], NSGA-II [24], TS-GEP [25], etc., but more scholars use models obtained from response surfaces or directly fitted models.

Meanwhile, to solve the problem that NSGA-II introduces fast non-dominated sorting method and elite strategy, defines crowding degree instead of adaptation value sharing, which reduces the computational complexity of the algorithm, but the algorithm has poor global search ability and is easy to fall into local optimum [26, 27]. The Gaussian distribution crossover operator is introduced to improve the search ability of NSGA, and the fast nondominated ranking genetic algorithm with elite strategy introducing Gaussian distribution crossover operator (GNSGA-II) is proposed for process parameter search.

Based on the established random forest regression prediction model, the Pareto solution set is obtained by using GNSGA-II for process parameter search with the quality characteristics and mechanical properties of the coating set as the objective, combined with engineering experience and the process range of equipment parameters as the constraints. Finally, laser cladding tests were conducted with 12Cr13 as the base material and 15-5PH as the cladding powder to verify the feasibility of the proposed GA-ACO-RFR predictive regression model combined with the GNSGA II multi-objective optimization algorithm approach. The optimal process parameters were also selected from the Pareto solution set, and validation tests were conducted to determine the effectiveness of the optimal process parameters obtained by optimization. It provides a fast and accurate method for obtaining laser cladding process parameters, which will help to promote intelligent and sustainable development of additive manufacturing.

2 Prediction Model and Optimization Method

The portfolio of laser cladding process parameters determines the quality characteristics of the coating, but the relationship between process parameters and quality characteristics is non-linear due to the interaction between process parameters. The interaction relationship is difficult to obtain with simple statistical methods and is difficult to apply in practical production. At the same time, there are also complex interactions between the quality characteristics of coatings. It is difficult to coordinate simple numerical analysis and single-objective optimization methods in order to meet different quality characteristics requirements. With the development of artificial intelligence, machine learning algorithms gradually show their unique superiority in multivariate nonlinear regression and multi-objective optimization problems.

Based on hybrid Genetic Algorithm and Ant Colony Optimization Random Forest Regression prediction algorithm (GA-ACO-RFR) and a fast non-dominated ranking genetic algorithm with an elite strategy of Gaussian distribution crossover operator (GNSGA-II) for predictive modeling and optimization of the quality characteristics of laser cladding coating to provide a reference set of

solutions for process parameters for surface modification of parts, as shown in Fig. 1.

The orthogonal experimental design is an economical, efficient and fast method to obtain the influence of each process parameter on the coating quality characteristics of laser cladding [12]. Based on the range of empirical process parameters, design of orthogonal tests on the parameters of the laser cladding process, preparation and data acquisition of a single coat using the laser cladding system, specimen processing equipment, and coating quality characteristics measurement equipment.

The hybrid Genetic Algorithm and Ant Colony Optimization algorithm (GA-ACO) is used to find the optimum random forest hyperparameters to obtain the optimum number of trees (*ntree*) and the number of randomly selected features (*ntry*). The test data of coating quality characteristics collected from orthogonal testing were used to create a Random Forest Regression prediction model (RFR) based on the ideal hyperparameters, with laser cladding process parameters as input and coating quality characteristics as output.

Finally, based on the prediction model established by the hybrid Genetic Algorithm and Ant Colony Optimization Random Forest Regression prediction algorithm (GA-ACO-RFR), a fast non-dominated ranking genetic algorithm with an elite strategy of Gaussian distribution crossover operator (GNSGA-II) is used to obtain a multi-objective optimization of the laser cladding process parameters with the quality

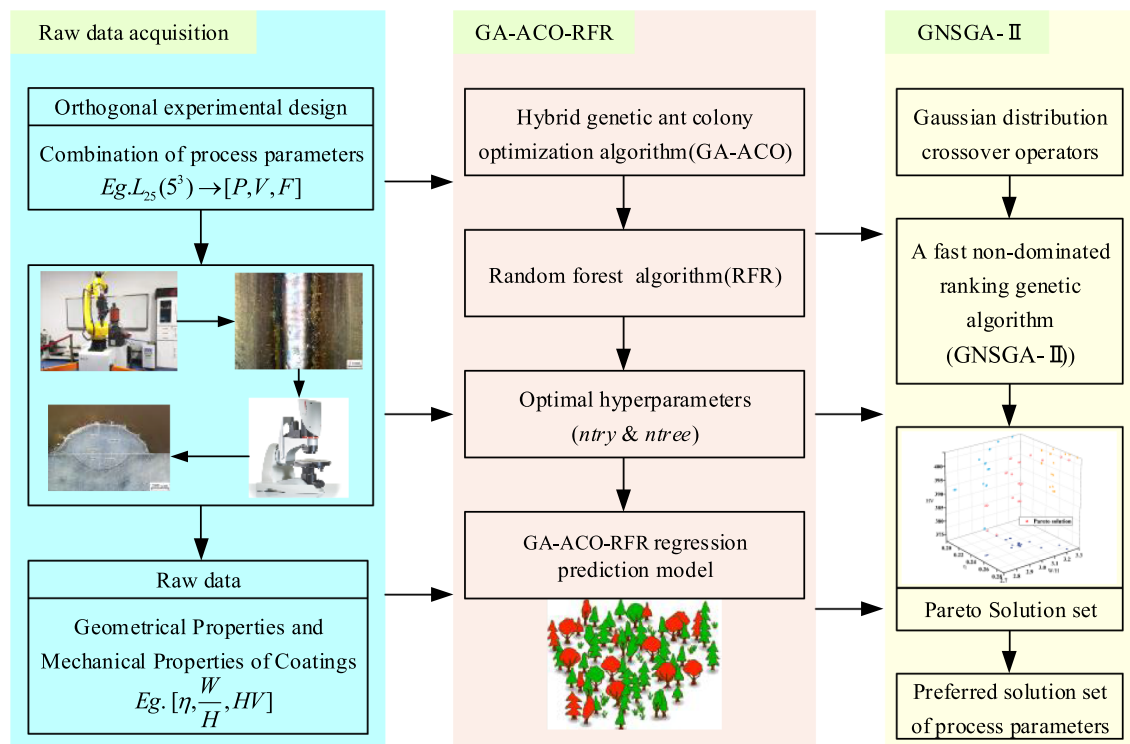


Fig. 1 Predictive modeling and optimization of laser cladding process parameters

characteristics of the coating as the objective, and a preferred solution set of the laser cladding process parameters is obtained.

2.1 Prediction Model of Cladding Layer Based on GA-ACO-RFR

The laser cladding process is influenced by the interaction of high-temperature thermal cycles and thermophysical properties of materials. Its forming mechanism is complicated and variable, and there are several process parameters that have an impact on the quality attributes. The random forest technique is one of the finest nonparametric regression models for prediction since it does not need testing the assumptions of normality and independence of variables or taking into account the co-linearity of many variables. It is capable of modeling the mapping between the quality characteristics and the laser cladding process parameters.

The random forest algorithm is a new type of machine learning algorithm proposed by *Breiman* in 2001 by combining the *Bagging* ensemble algorithm and the *CART* decision tree algorithm [28]. A hybrid Genetic Algorithm and Ant Colony Optimization algorithm (GA-ACO) [29] was used to optimize the hyperparameters of the Random Forest Regression algorithm (RFR) to obtain the optimum random forest performance. The Random Forest Regression prediction model of laser cladding process parameters and coating quality characteristics was constructed based on the optimal hyperparameters, to realize the establishment of a highly mapping model between coating quality and process parameters. Flow chart of GA-ACO-RFR regression prediction model, as shown in Fig. 2.

2.1.1 Random Forest Regression Prediction Model

The input matrix X is composed of a portfolio of laser cladding process parameters, with M features (laser power P , scanning speed V , powder feeding speed F , etc.) and N samples (the number of designed orthogonal experiment groups), the quality characteristics of the coating form the objective vector Y , which constitutes the training data set $S = \{x_i, y_i\}$, $(X, Y) \in R^M * R$.

A sub-training sample (S_k ($k = 1, 2, \dots, ntree$)) with the same size as the original training sample is extracted from the original training sample by the playback sampling method (Bootstrap resampling). Each sub-training sample constitutes a decision tree. When building a decision tree, each non-leaf node randomly selects $ntry$ feature split nodes from M features. Finally, a classification regression algorithm (*CART*) is used to select the optimal splitting method training from the randomly selected features, and the mean value of all decision tree prediction results is used as the

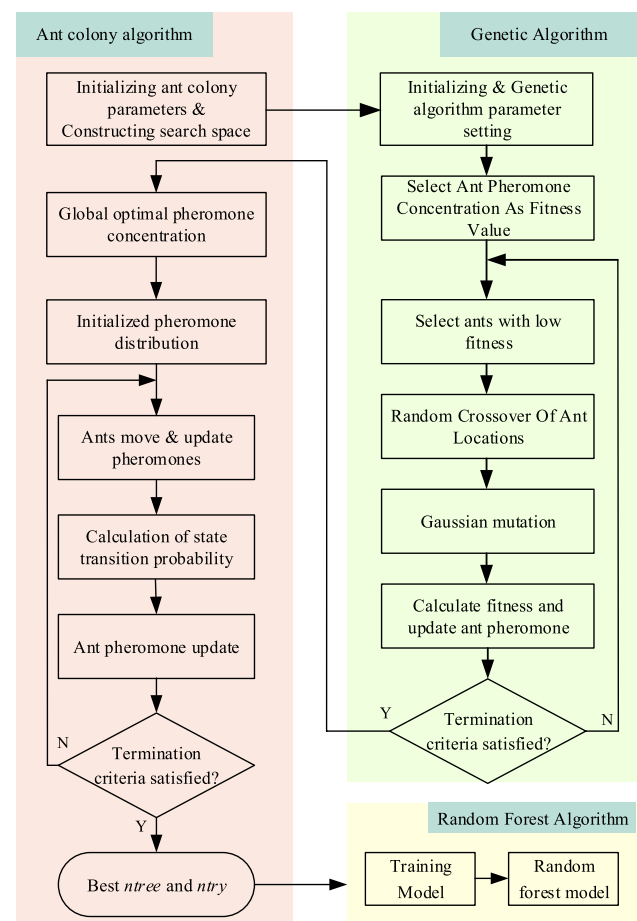


Fig. 2 Flow chart of GA-ACO-RFR

final prediction result [30]. The structure of the random forest regression algorithm is shown in Fig. 3.

The number of trees in the random forest $ntree$ and the number of random screening features $ntry$ determine the performance of the random forest when constructing RFR. The random forest model performs poorly when the $ntree$ is small and each decision tree produces distinct random decisions that are entirely independent of one another. The diversity of base learners likewise rises as $ntree$ increases, but the interpretability of the model decreases as calculation and memory requirements rise [31].

When creating a single decision tree, the random forest first randomly chooses screening features at the split node position from the candidate feature set of the first training sample. It next chooses the optimal splitting method from the screening features. The degree of randomization and prediction accuracy of the model decrease with decreasing $ntry$, while the variance of the model increases with increasing $ntry$ [32]. The conventional approach, which calls for a lot of resources and labor and has a low efficiency, makes it challenging to determine the ideal random forest parameters.

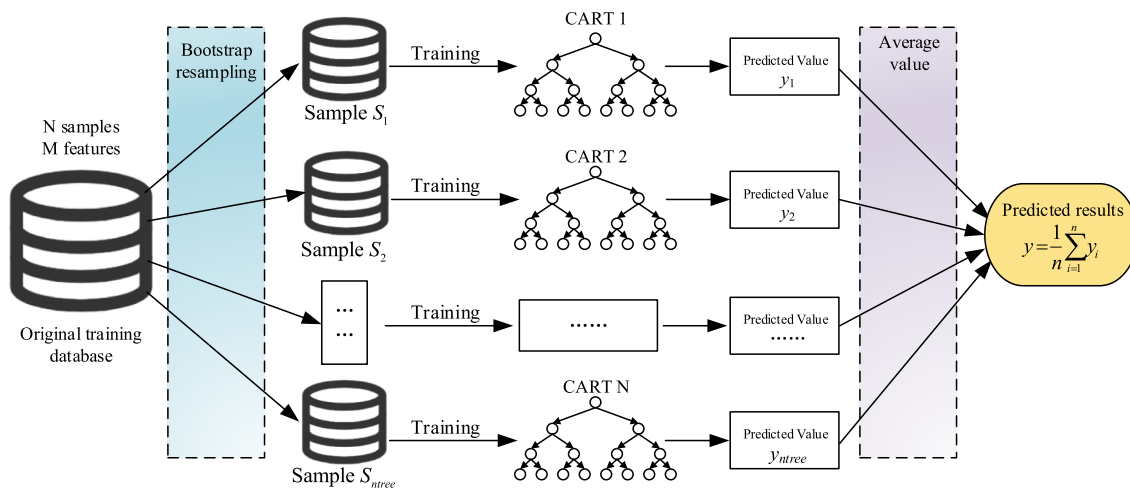


Fig. 3 Schematic diagram of random forest

2.1.2 Hybrid Genetic Algorithm and Ant Colony Optimization Random Forest Regression Algorithm Hyperparameters

Genetic Algorithm is an adaptive random iterative search method. The solution of the problem is gradually optimized and the optimal solution is obtained through repeated evolution. The crossover factor and mutation operator make it likely that every solution in the population will be examined, expanding the scope of the global search and not easily falling into local optimal solutions. Despite having a quick and global search capability, GA has a low solution efficiency [33].

Ant colony algorithm is a heuristic search algorithm based on population optimization [34]. Its search method uses a distributed computing architecture, which boosts the algorithm's efficiency and computation speed. The search is gradually brought into focus and eventually approaches the ideal solution thanks to the positive feedback mechanism. Positive feedback is simple to fall into the local optimum, nonetheless, because of the pheromone concentration disposal's uncertainty. The ACO algorithm has good parallel solution quality and strong robustness, but it also has drawbacks including sluggish convergence and easy local optimum fall-in. A hybrid Genetic Algorithm and Ant Colony Optimization algorithm is created by combining the advantages of the genetic algorithm and ant colony algorithm.

The hybrid Genetic Algorithm and Ant Colony Optimization algorithm uses GA and ACO to guide each other during each cycle by updating the global optimal solution. The optimal solution obtained by the ACO algorithm is added to GA as an elite individual to improve the accuracy and efficiency of GA in solving problems. To update the pheromone concentration of the ACO algorithm and enhance its overall optimization efficiency, the optimal

solution produced by GA is employed as the algorithm's current optimal location. The hybrid Genetic Algorithm and Ant Colony Optimization algorithm obtains the advantages of GA and ACO through multiple cycles to avoid getting into local optimal solutions and speed up the process of finding optimal solutions [29].

The GA-ACO is used to optimize the decision tree number $ntree$ and split feature number $ntry$ of the RFR model to improve the output accuracy of the RFR model. Iterative search for optimal hyperparameters of the random forest by GA-ACO, and the optimal solution obtained by the GA-ACO is used as the number of decision trees $ntree$ and the number of split features $ntry$ of the RFR model to obtain the optimum random forest regression prediction model.

The specific steps of the GA-ACO-RFR model are as follows:

- (1) Initialization. According to the optimization problem to be solved, initialize the genetic algorithm and set the selection method, crossover probability, variation probability, number of iterations, etc. Initialize the number of ants in the ant colony algorithm, the pheromone volatilization coefficient and the objective error.
- (2) Select the fitness function. The ant pheromone concentration is used as the fitness value of the genetic algorithm.
- (3) Construct the solution space and update the ant pheromone. The initial position of the ants is randomly set, and S ants start to search and update the pheromone. At the same time, the ants update the pheromone value of the path they pass through in real-time according to formula (1). Sort the updated ant pheromone in ascending order to form the solution space.

$$\begin{cases} \pi_j(I_{ni}) = (1 - \rho)\pi_j(I_{ni}) + \Delta\pi_j(I_{ni}) \\ \pi_j(I_{ni}) = \sum_{k=1}^m \Delta\tau_j^k(I_{ni}) \end{cases} \quad (1)$$

where ρ is the pheromone volatility coefficient; $\Delta\tau_j^k(I_{ni})$ is the amount of information on the path of element j of the k_{th} ant in the current cyclic set.

- (4) Introduce a genetic algorithm to perform genetic selection, crossover, mutation and other operations on ant colonies. The ants are sorted in ascending order of fitness, and a certain percentage of ants with low pheromones are taken for genetic selection, crossover and mutation. First, a linear crossover is adopted. For two chromosomes x_i and $x_j(i=j)$, the linear crossover is shown in formula (2).

$$x'_{id} = \lambda \cdot x_{id} + (1 - \lambda) \cdot x_{jd}, i = 1, 2, \dots, N; d = 1, 2, \dots, D \quad (2)$$

where λ is a random number on the interval of $[0, 1]$.

Then, Gaussian mutation is performed. For a chromosome x_i , the mutation individual is:

$$x_{id} \leftarrow G(x_{id}, \frac{1}{n}), d = 1, 2, \dots, D \quad (3)$$

- (5) The iteration starts to obtain the global optimal solution. Determine whether the maximum number of iterations is reached. If the maximum number of iterations is reached, output the global optimal solution and proceed to the next step. Otherwise, the loop is only step4 until the maximum number of iterations is reached.
- (6) Initialize the ACO algorithm pheromone. Initialize the pheromone of the ACO algorithm using the optimal solution obtained by step5.
- (7) Calculate transfer probability. Calculate the state transition probability according to the formula (4).

$$p_{ij}^k(t) = \begin{cases} \frac{[\tau_{ij}(t)]^\alpha [\eta_{ij}(t)]^\beta}{\sum_{s \in J_k(i)} [\tau_{is}(t)]^\alpha [\eta_{is}(t)]^\beta}, j \in J_k(i) \\ 0, \text{ else} \end{cases} \quad (4)$$

The probability ($p_{ij}^k(t)$) of ant k walking to city j in city i at time t . Where n is the number of cities, m is the number of ants, $d_{ij}(i, j = 1, 2, \dots, n)$ is the distance of city i and j . $\tau_{ij}(t)$ is the pheromone concentration on route i, j in the city at time t , k is the path chosen by the ants according to the pheromone concentration. $J_k(i) = [1, 2, \dots, n - 1] - \text{tabu}_k$ is the set of ant k cities to be transferred.

- (8) Update the pheromone of the ant. The pheromone concentration of the k_{th} ant in the i_{th} subpopulation was

updated according to the pheromone update formula (5).

$$\tau(i, j) = (1 - \rho) \times \tau(i, j) + \sum_{k=1}^m \Delta\tau_k(i, j) \quad (5)$$

- (9) Iterate to obtain the global optimal solution. Compare the obtained optimal solutions, and update the global optimal solution if it is optimal; otherwise, loop step7 ~ step8 until the optimal solution is obtained or the maximum number of iterations is reached.
- (10) Random forest model construction. The global optimal solution obtained from step9 is used to construct the random forest model and train it.

2.1.3 The Evaluation Metrics of Model

The GA-ACO-RFR laser cladding prediction model was established mainly to find the optimal process parameters. To ensure the accuracy of the established prediction model, the performance of the random forest prediction model was checked and evaluated. The following indicators were selected for evaluation:

- (1) *MAPE*

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - f(x_i)}{y_i} \right| \times 100\% \quad (6)$$

where n is the number of predicted samples, y_i is the true value, $f(x_i)$ is the predicted value of the model.

MAPE is a strong indicator of the performance of predictive models for laser cladding geometric quality characteristics and mechanical properties. The smaller the *MAPE* indicates that the prediction error of the model is smaller, the higher the prediction accuracy of the model, and the better the performance of the model.

- (2) *MSE*

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 \quad (7)$$

where n is the number of predicted samples, y_i is the true value, $f(x_i)$ is the predicted value of the model.

The *MSE* is a reflection of the dispersion of the sample data, and a lower value indicates that the prediction error of the laser cladding geometric quality characteristics and mechanical properties prediction model is smaller.

- (3) R^2

$$R^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})(f(x_i) - \bar{f})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (f(x_i) - \bar{f})^2}} \quad (8)$$

where n is the number of predicted samples, y_i is the true value, $f(x_i)$ is the predicted value of the model.

R^2 is a statistical measure that reflects the linear correlation between the true and predicted values. The higher the correlation coefficient, the better the predicted results of the geometric and mechanical properties of laser cladding fit the true values, and the stronger the interpretability of the model, which usually takes values between 0 and 1.

(4) Iterative comparison of algorithms

The fitness function selects the root mean square error average of the training and test sets as a whole, the formula for calculating the fitness function is shown in Eq. (9), and the smaller its value the better the accuracy and prediction precision of the model. At the meantime, the smaller the number of iterations of the algorithm indicates the faster the convergence of the algorithm and its better spatial search capability.

$$fitness = \frac{mse(out_train, train_simu) + mse(out_test, test_simu)}{2} \quad (9)$$

where $mse()$ denotes the mean square error of taking two values; out_train denotes the true output value of the training set and $train_simu$ denotes the predicted output value of the training set; out_test denotes the true output value of the test set and $test_simu$ denotes the predicted output value of the test set.

2.2 Multi-objective Optimization of Laser Cladding Process Parameters Based on GNSGA-II

2.2.1 Definition of Quality Characteristics of Laser Clad Coatings

To obtain the optimal laser cladding process parameters, the laser cladding process parameters need to be determined with the quality characteristics of the coating as the optimization objective. According to the engineering experience, the objective ratio of coating width and height $W/H=3$ is chosen for the subsequent preparation of multilayer multi-layer coating with excellent surface flatness [35]. Too low dilution rate belongs to mechanical occlusion, and the bonding force is poor or even cannot form metallurgical bonding; too high a dilution rate is likely to cause excessive dilution of the coating by the substrate,

increasing the tendency of cracking and deformation, resulting in low coating performance. So, the objective dilution rate $\eta=25\%$ is chosen here [36]. To obtain reliable performance of the coatings, the greater the microhardness of the coatings, the better. The optimization objective is shown in Eq. (10).

$$\begin{cases} \min |f_{W/H}(P, V, F) - 3| \\ \min |f_{\eta}(P, V, F) - 25\%| \\ \max f_{HV}(P, V, F) \end{cases} \quad (10)$$

2.2.2 The Fast Non-Dominated Ranking Genetic Algorithm with Elite Strategy by Introducing the Gaussian Distribution Crossover Operator (GNSGA-II)

Since the optimization of laser cladding process parameters is a multi-objective coupled nonlinear optimization process, a fast non-dominated ranking genetic algorithm with an elite strategy introducing a Gaussian distribution crossover operator (GNSGA-II) is used to optimize the laser cladding process parameters, as shown in Fig. 4.

NSGA-II is a fast non-dominated ranking genetic algorithm with an elite strategy, which can achieve fast and accurate search and incorporates an elite strategy to ensure that the optimal solution found will not be discarded [37, 38]. However, the algorithm has poor global search ability and is prone to fall into local optimal solutions. The Gaussian distribution crossover operator is introduced to improve the search capability of NSGA-II. In this paper, the improved algorithm names GNSGA-II.

The evolutionary operation of NSGA-II uses analog binary crossover [27], the basic process of crossover is as follows: (1) Two individuals were randomly selected from the population obtained by the selection operation; (2) Generate a distributed random number; (3) Calculation of parameters by polynomial probability distribution; (4) Calculating crossover progeny, as shown in Eq. (11).

$$\begin{cases} c_1 = \frac{p_1 + p_2 - \beta \cdot |p_2 - p_1|}{2} \\ c_2 = \frac{p_1 + p_2 + \beta \cdot |p_2 - p_1|}{2} \end{cases} \quad (11)$$

where c_1 & c_2 is the child individual, p_1 & p_2 is the parent individual, and β is the random variable.

The Gaussian distribution crossover operator is introduced into the crossover operation and the discrete reorganization of the evolutionary strategy is performed [39], the final Gaussian distribution crossover operator is established as follows:

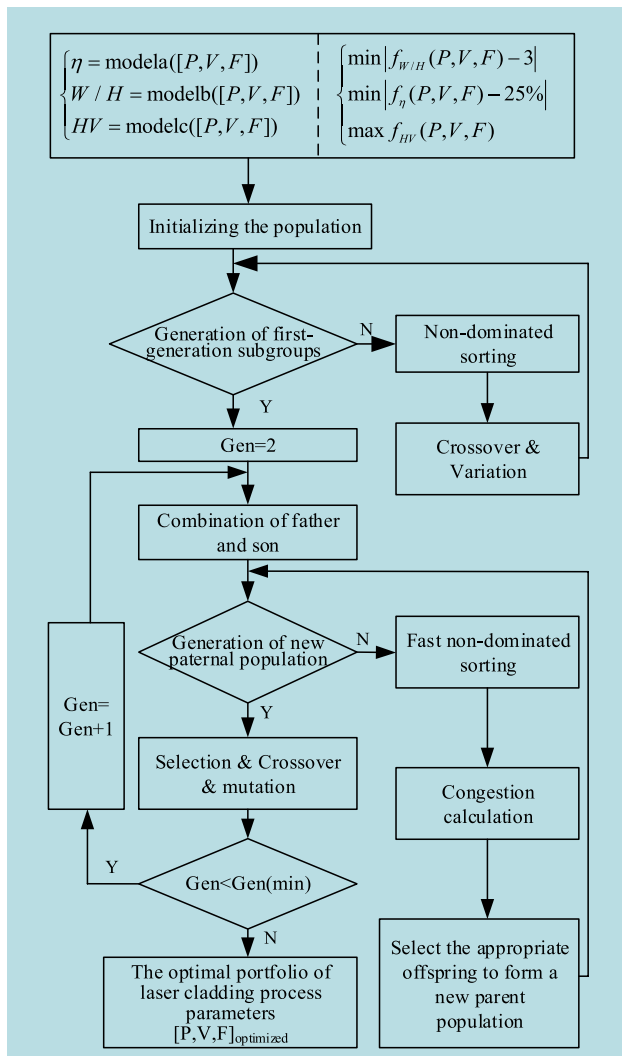


Fig. 4 Flow chart of GNSGA-II multi-objective optimization algorithm

$$\begin{cases} c_1 = \frac{p_1 + p_2 + 1.481 \cdot (p_1 - p_2) \cdot N}{2} \alpha \leq 0.5 \\ c_2 = \frac{p_1 + p_2 - 1.481 \cdot (p_1 - p_2) \cdot N}{2} \alpha \leq 0.5 \\ c_1 = \frac{p_1 + p_2 - 1.481 \cdot (p_1 - p_2) \cdot N}{2} \alpha > 0.5 \\ c_2 = \frac{p_1 + p_2 + 1.481 \cdot (p_1 - p_2) \cdot N}{2} \alpha > 0.5 \end{cases} \quad (12)$$

where N is a Gaussian distributed random variable, α is a random number uniformly distributed in $(0,1)$, $A = 1.481$ is obtained from the Eq. (13).

$$\int_{-1/A}^{1/A} \frac{1}{2\pi} e^{-x^2/2} dx = 0.5 \quad (13)$$

The Gaussian distribution crossover operator is introduced into the NSGA-II algorithm to expand the spatial search range, avoid generating local optima and instability

of the evolutionary process, and ensure the integrity and normal distributivity of the Pareto solution set.

Finally, based on the established GA-ACO-RFR regression prediction model, the process parameters were optimized by combining with GNSGA-II multi-objective optimization algorithm to obtain the set of process parameters satisfying the constraints.

2.2.3 Multi-objective Optimization Effect Evaluation Index

Hypervolume [40, 41] is used to measure the volume of the target space, the volume of the region of the target space enclosed by the non-dominated solution set and the reference points. Larger *Hypervolume* values indicate better convergence and diversity of the algorithm.

$$Hypervolume = \delta \left(\bigcup_{i=1}^{|S|} v_i \right) \quad (14)$$

Hypervolume represents the Lebesgue measure, which is used to measure the volume. $|S|$ denotes the number of non-dominated solution sets, and v_i denotes the *Hypervolume* formed by the reference point and the i -th solution in the solution set.

The algorithm running time is chosen as a metric to evaluate the convergence speed. The shorter the overall running time of the algorithm, the better the algorithm's ability to find the best.

3 Experiment

3.1 Preparation of the Coatings

The substrate material was 12Cr13 hot-rolled stainless steel plate. The surface of the raw material was ground by a surface grinder to remove the oxide layer and cut into specimens of size $60 \times 55 \times 10$ mm, cleaned and dried with ethanol before the experiment. The selected clad coating powder is 15-5PH spherical alloy powder shown in Fig. 5, with a particle size of $48\text{--}120$ μm . The 15-5PH powder is dried in an oven at 200 $^{\circ}\text{C}$ for 2 h before the experiment to ensure the fluidity of the powder and avoid defects such as cracks and pores in the coating. 12Cr13 stainless steel and 15-5PH alloy powder chemical composition is shown in Table 1.

The laser cladding system consists of a coaxial four-way powder feeding laser cladding head, multi-functional powder feeder, continuous fiber laser generator, work table, etc. The cladding process is protected from oxidation by a high purity argon atmosphere. The laser cladding system is shown in Fig. 6, and the relevant parameters are shown in Table 2.

The quality of single-track laser cladding coating is mainly determined by the controllable parameters of the laser cladding system. The controllable process parameters in the laser

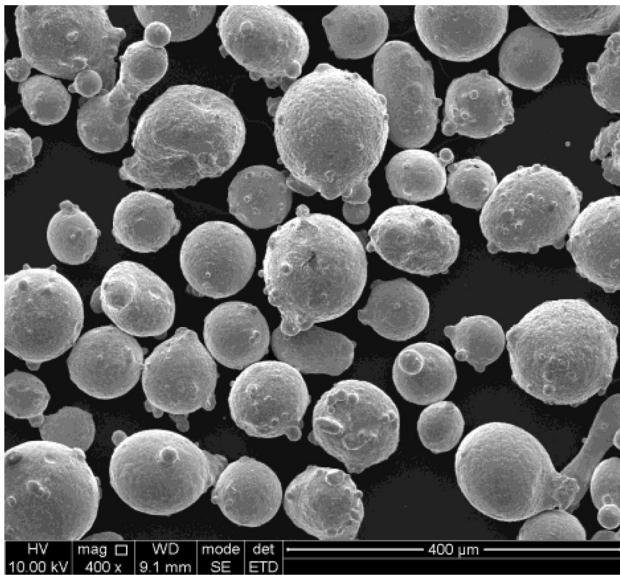


Fig. 5 The spherical diagram of 15-5PH powder

cladding system are mainly composed of laser power, powder feeding speed, scanning speed, spot diameter, and defocusing amount. Based on the previous research experience, the laser power, powder feed speed, and scanning speed, which are often adjusted in engineering, are selected as the research parameters in this study. The adjustment range is selected according to the engineering experience to form the input parameter matrix $X = \begin{bmatrix} P_1 & V_1 & F_1 \\ \vdots & \vdots & \vdots \\ P_N & V_N & F_N \end{bmatrix}$. The rest of the process parameters are fixed according to the empirical values.

The characteristic data of single-track laser cladding coating were obtained, the laser spot was set to 3 mm, the protective gas flow rate was 10 L/min, and a three-factor five-level ($L_{25}(5^3)$) orthogonal experiment was designed in the range of engineering empirical values of laser power, powder feeding speed, and scanning speed, as shown in Table 3.

The single-track laser cladding length was set to 45 mm, the single-track clad specimens are shown in Fig. 7. The surface of the clad channel was observed using a Leica DVM6S ultra-deep field microscope and analyzed by ultra-deep field stacking, and it was found that the surface of the clad channel was smooth and no cracks were found. The specimens were cut into $10 \times 10 \times 10$ mm specimens using a metallographic cutter, sanded with 200 to 1200# sandpaper

after inlaying, then polished using a grinding and polishing paste combined with an automatic grinding and polishing machine, and finally rinsed and dried with anhydrous ethanol. A mixed etching solution with the volume ratio of $\text{HNO}_3:\text{CH}_3\text{COOH}:\text{CH}_3\text{CH}_2\text{OH}=2:3:3$ was prepared, and the specimens were wiped and etched with cotton swabs dipped in the etching solution until the coating boundary was clearly shown. Observation found that the fusion channel cross-section shows that there is an obvious condensation line between the coating and the substrate, indicating that the coating and the substrate formed a metallurgical bond, the coating, and the substrate are not found to be inclusions, pores, cracks, and other defects.

The HV-1000B microhardness tester was used to test the cross-sectional hardness under the conditions of loading 500 g and holding pressure for 10 s. The schematic diagram of the measurement points is shown in Fig. 8, and the microhardness at 5 locations of the coating was collected and the average value was taken as the microhardness value of the coating.

3.2 Indicator Definitions and Characterization

After corrosion, the specimens were measured by ultra-deep field microscope for the coating clad width W , clad height H , clad depth d_m , coating area A_c and clad pool area A_m , and the width to height ratio W/H and dilution rate η was

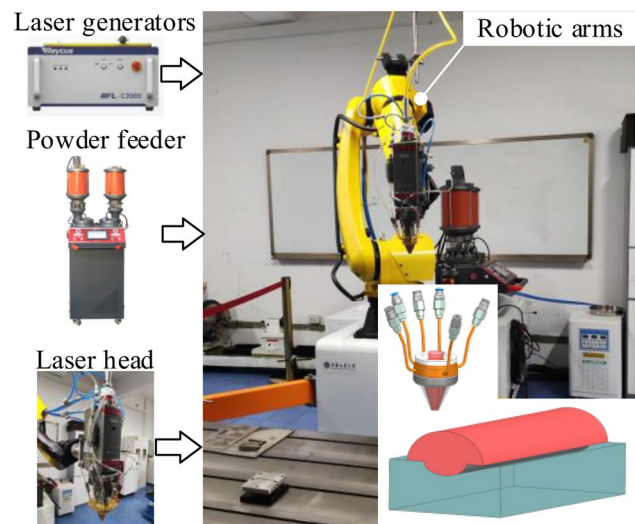


Fig. 6 Laser cladding system

Table 1 Chemical composition of 12Cr13 and 15-5PH

Material	Elements wt(%)									
	C	Mn	Si	P	S	Cr	Ni	Cu	Nb	Fe
12Cr13	0.14	0.25	0.42	0.44	0.03	12.05	0.05	—	—	Bal
15-5PH	0.03	0.84	0.38	0.01	0.01	15.41	4.66	3.71	0.3	Bal

Table 2 Basic information of the laser cladding equipment

Device model	Items	Values
Laser generators(RFL-C3000)	Range of laser power(W)	0–3000
	Center wavelength(nm)	1080±5
	Power range(%)	10–100
	Modulation frequency (kHz)	5
	Optical fiber core diameter (μm)	20
Laser printheads(TS-6.0A)	Range of power(kW)	6–15
	Powder focus diameter(mm)	2–6
	Horizontal adjustment range(mm)	±30
	Collimated focal length(mm)	100
Powder feeder(ECPF-DCM-D1-2-P2)	Powder feeding amount(g/min)	4–200(±1%)
	Powder disc speed(r/min)	0–16
	Air pressure(bar)	2–6
	powder particles(μm)	20–250
	air flow volume(L/min)	1–25

Table 3 Experiment design table

NO	P(W)	V(mm/s)	F(r/min)	NO	P(W)	V(mm/s)	F(r/min)
1	1300	9	0.9	14	1500	12	0.9
2	1300	10	1	15	1500	13	1
3	1300	11	1.1	16	1600	9	1.2
4	1300	12	1.2	17	1600	10	1.3
5	1300	13	1.3	18	1600	11	0.9
6	1400	9	1	19	1600	12	1
7	1400	10	1.1	20	1600	13	1.1
8	1400	11	1.2	21	1700	9	1.3
9	1400	12	1.3	22	1700	10	0.9
10	1400	13	0.9	23	1700	11	1
11	1500	9	1.1	24	1700	12	1.1
12	1500	10	1.2	25	1700	13	1.2
13	1500	11	1.3	–	–	–	–

calculated by Eq. (15) and Eq. (16) respectively, and the schematic diagram of the coating cross-section is shown in Fig. 8. The obtained coating dilution rate, width-to-height ratio, and average microhardness form the objective matrix

$$Y = \begin{bmatrix} \eta_1 & (W/H)_1 & HV_1 \\ \vdots & \vdots & \vdots \\ \eta_N & (W/H)_N & HV_N \end{bmatrix}.$$

$$\eta = \frac{A_m}{A_m + A_c} \quad (15)$$

$$W/H = \frac{W}{H} \quad (16)$$

3.3 Prediction and Optimization Process

Coating dilution rate, width-to-height ratio and measured average microhardness were calculated using the formulae

related to coating geometric quality characteristics and mechanical properties to form the data set, and 80% of the data were divided to form the training set to train the GA-ACO-RFR prediction model to obtain the random forest regression prediction model. The parameters of the genetic algorithm were set as follows: the population size was set to 50, the maximum number of iterations was 150, the crossover probability was 0.9, and the variation probability was 0.01 to satisfy the maximum number of iterations to end the evolution. The parameters of the ant colony algorithm were selected as the number of ants 10 and the pheromone volatility coefficient 0.6. The decision tree *ntree* in the random forest was taken in the range of [10,500] and the number of random screening features *ntry* was taken in the range of [1, 9]. The obtained data were substituted into the data to construct GA-ACO-RFR regression prediction models of process parameters regarding coating dilution rate, width-to-height ratio, and average microhardness, respectively.

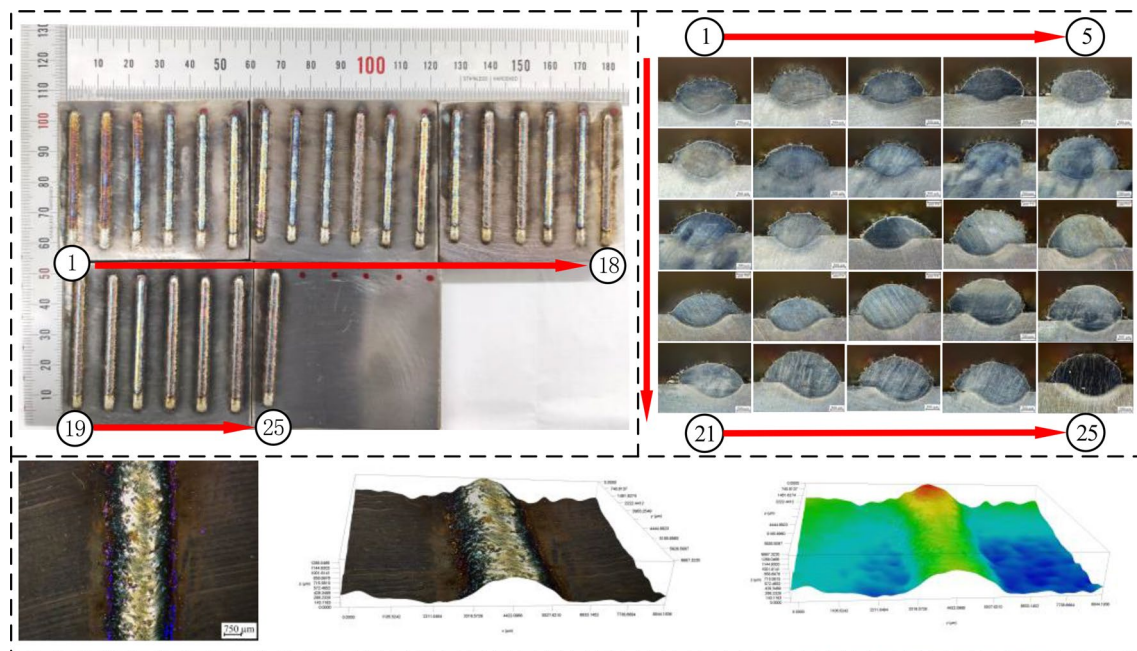


Fig. 7 Macroscopic morphology of the experimental cladding layer

The established random forest regression prediction model for laser cladding coating dilution rate, width-to-height ratio and average microhardness is set as the objective function of the GNSGA II multi-objective optimization algorithm, as shown in Eq. (17). For laser cladding powders with unknown forming properties and geometric quality characteristics, the engineering experience combined with the equipment parameters is used to extend the process parameter range to prevent the wrong selection of the optimal process parameters due to the lack of engineering experience, and Eq. (12) is the extended process parameter range. The extended range of laser cladding process parameters is used as a constraint as shown in Eq. (10). The parameters

of the GNSGA II multi-objective optimization algorithm are set as follows: evolutionary generation 100, number of populations 150, crossover probability 0.9, and variation probability 0.01.

$$\begin{cases} 1100\text{W} \leq P \leq 2000\text{W} \\ 7\text{mm/min} \leq V \leq 15\text{mm/min} \\ 0.7\text{r/min} \leq F \leq 1.5\text{r/min} \end{cases} \quad (17)$$

4 Results and Discussion

4.1 Result Analysis and Validation of Prediction Model

The comparison graph of convergence between GA-ACO-RFR algorithm and ACO-RFR algorithm is shown in Fig. 9. The GA-ACO-RFR algorithm converges at the 7th generation of iteration and the ACO-RFR algorithm converges at the 15th generation of iteration. Obviously, the GA-ACO-RFR algorithm converges faster. And the fitness value at convergence of GA-ACO-RFR algorithm is smaller than that of ACO-RFR algorithm, which indicates the better prediction accuracy and precision of GA-ACO-RFR algorithm. The main reason is that the introduction of the crossover operator and variation operator of the genetic algorithm in the ant colony algorithm expands the search space of the ant colony algorithm, avoids getting into local

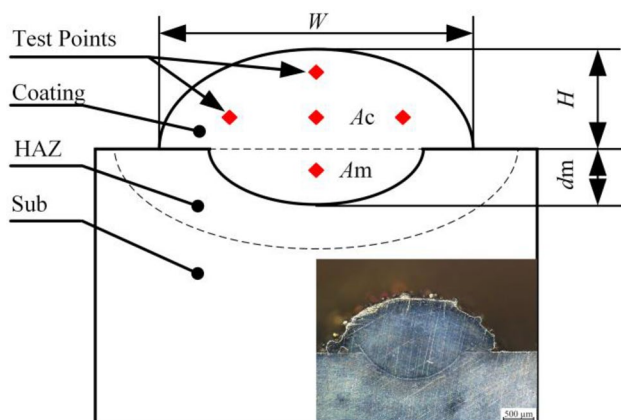


Fig. 8 Schematic diagram of the cross-section morphology and microhardness measuring points of cladding sample

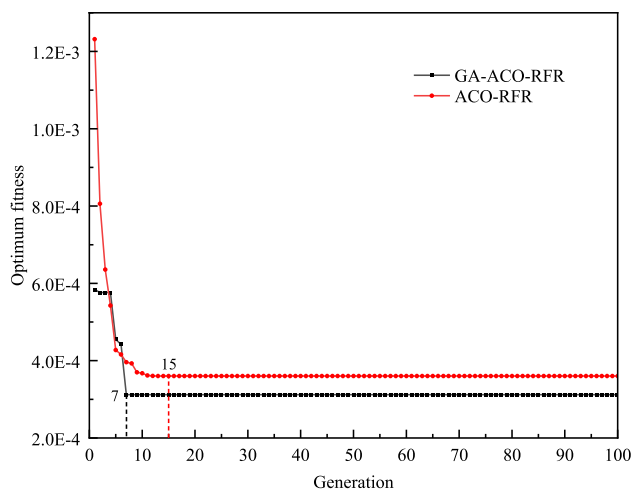


Fig. 9 Iterative comparison of two algorithms

optimum, and improves the search ability and convergence speed of the ant colony algorithm.

To illustrate the effectiveness of the GA-ACO algorithm improved prediction RFR model, the results of the GA-ACO-RFR prediction model were compared with the unoptimized RFR model by setting the number of decision trees and random screening features in RFR to values within a randomly selected range of values, respectively. As shown in Fig. 10, the comparison between the GA-ACO-RFR prediction model and the RFR prediction model for each tested value and the true value shows that the data tested by the GA-ACO-RFR prediction model are closer

to the true value than those obtained by the RFR prediction model.

Combining the evaluation indexes of the two models in Table 4, the evaluation indexes MAPE and MSE of each output of the GA-ACO-RFR prediction model are lower than those of the unoptimized RFR prediction model, indicating that the GA-ACO-RFR algorithm has good fitting effect and accuracy, and the deviation between the predicted and true values is small. The Pearson correlation coefficient (R^2) is higher and close to 1 than that of the RFR prediction model, indicating that the GA-ACO-RFR prediction model has low prediction error, high prediction accuracy, high fit of the prediction results of coating geometric quality characteristics and average microhardness to the true values, and strong interpretability of the model.

As shown in Table 5, three sets of laser cladding process parameter combinations before (NO.4-NO.5) and after (NO.1-NO.3) the extension were randomly selected for repeatability validation experiments, respectively, to verify the prediction ability of the GA-ACO-RFR prediction model for the out-of-training range of the model and the repeatability of the experiments. The results are shown in Table 6. The errors between the predicted and measured values are less than 5%, indicating that the prediction performance of the GA-ACO-RFR prediction model for global process parameters is stable and reliable. Combining the prediction model evaluation index and validation experiments, the optimized RFR prediction model using GA-ACO can accurately predict the geometric quality characteristics and average microhardness of the coating and can be used for

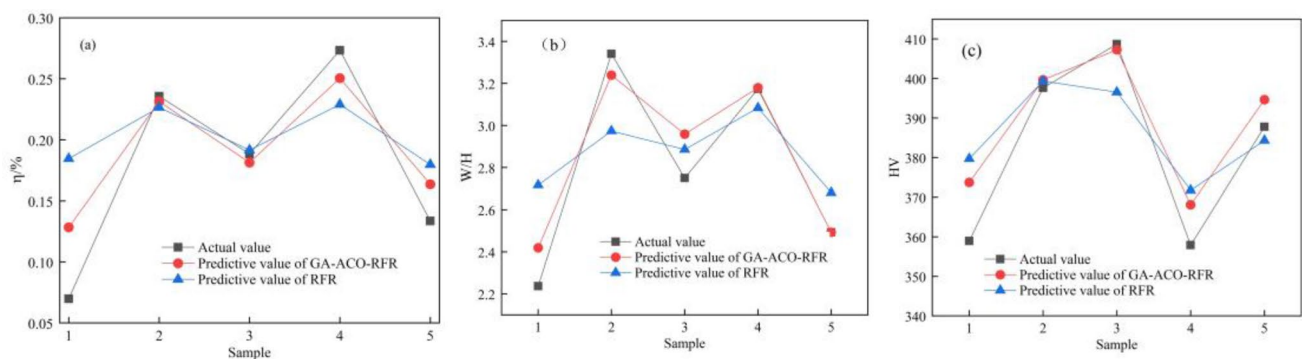


Fig. 10 The predicted value of the test set compared with the actual value in GA-ACO-RFR & RFR: **a** dilution rate, **b** W/H, **c** microhardness (HV)

Table 4 Quantitative representation of the modeling results of GA-ACO-RFR

Indicators	GA-ACO-RFR			RFR		
	MAPE/%	MSE	R^2	MAPE/%	MSE	R^2
η	15.75610	0.00098	0.97067	22.54500	0.00346	0.77653
W/H	3.58610	0.01746	0.93607	8.94040	0.08554	0.79669
HV	1.86000	74.76060	0.98598	2.71780	156.80240	0.83489

Table 5 Selected process parameters before and after expansion

NO	P(W)	V(mm/min)	F(r/min)	W(μ m)	H(μ m)	dm(μ m)	Ac(μ m ²)	Am(μ m ²)
1	1500	9	0.9	2854.72	999.84	499.84	2,066,561.32	691,556.43
2	1600	11	1	2511.91	941.38	466.27	1,980,988.28	574,916.89
3	1700	12	1.2	2954.46	1064.80	400.81	2,593,927.40	561,823.23
4	1100	14	0.8	1881.13	806.81	203.24	897,007.60	120,271.00
5	1800	8	0.7	2536.71	921.58	843.56	1,872,792.98	710,627.81
6	1900	15	1.5	2977.62	1136.18	507.71	2,603,942.43	601,944.95

Table 6 Validation results of selected process parameters before and after the extension

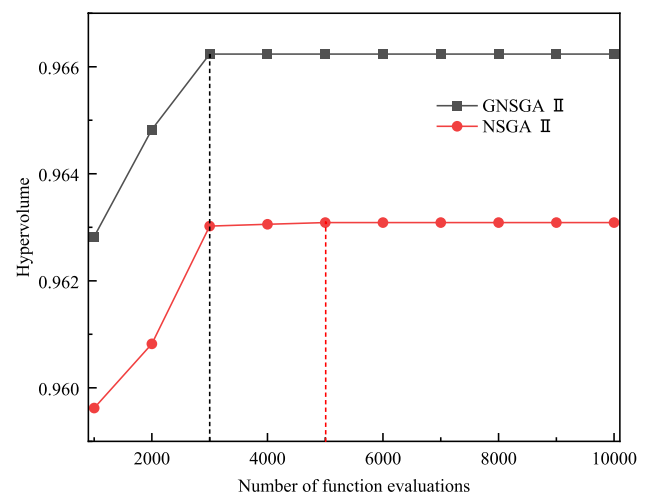
NO	η	W/H			HV			Validation	Error (%)
		GA-ACO-RFR	Validation	Error (%)	GA-ACO-RFR	Validation	Error (%)		
1	0.2507		0.2497	0.40	2.8552	2.8452	0.35	407.23	2.64
2	0.2249		0.2170	3.54	2.6683	2.7054	1.39	402.43	2.54
3	0.1780		0.1825	2.51	2.7747	2.7322	1.53	392.70	1.36
4	0.1182		0.1157	2.14	2.3316	2.2907	1.75	372.47	3.19
5	0.2751		0.2625	4.56	2.7526	2.7864	1.23	405.30	0.44
6	0.1878		0.1851	1.41	2.6207	2.6402	0.74	396.80	2.52

subsequent multi-objective optimization of the laser cladding process parameters.

4.2 Analysis and Validation of Optimization Results

As shown in Fig. 11, the super volume index of GNSGA-II is significantly larger than that of NSGA-II, which indicates that the convergence and diversity of GNSGA-II are significantly improved relative to NSGA-II. The average time of 10 repetitions of each independently, comparing and analyzing the solution speed of the two algorithms. the average time of 10 repetitions of GNSGA-II is 56.868 s, and the average running time of NSGA-II is 69.478 s. The solution speed of GNSGA-II algorithm is better than NSGA-II, further confirming the superiority of GNSGA-II.

The optimal solution set of the Pareto frontier was obtained by iterative optimization as shown in Fig. 12. The combination of process parameters of the Pareto optimized solution set obtained by GNSGA-II: laser power $P=1733$ W, scanning speed $V=11$ mm/s, and powder feeding speed $F=1.2$ r/min was selected as the optimum portfolio of process parameters. And the process parameters before optimization obtained from the orthogonal experimental results combined with engineering experience were selected as the reference group (laser power $P=1500$ W, scanning speed $V=12$ mm/s, powder feeding speed $F=0.9$ r/min). The comparison of coating geometric quality characteristics and average microhardness results before and after optimization is shown in Table 7. The coating dilution rate was increased by 44.71%, and the bonding strength of the coating and the substrate was improved; the width-to-height

**Fig. 11** Comparison of the hypervolume variation of the two algorithms

ratio was increased by 36.26%, which could pave the way for the subsequent acquisition of a better multi-pass fusion surface flatness; the average microhardness of the coating was increased by 12.71%, and the mechanical properties of the coating were strongly improved.

5 Conclusions

A regression prediction model between the laser cladding process parameters (laser power, scanning speed, powder feeding speed) and the coating quality characteristics

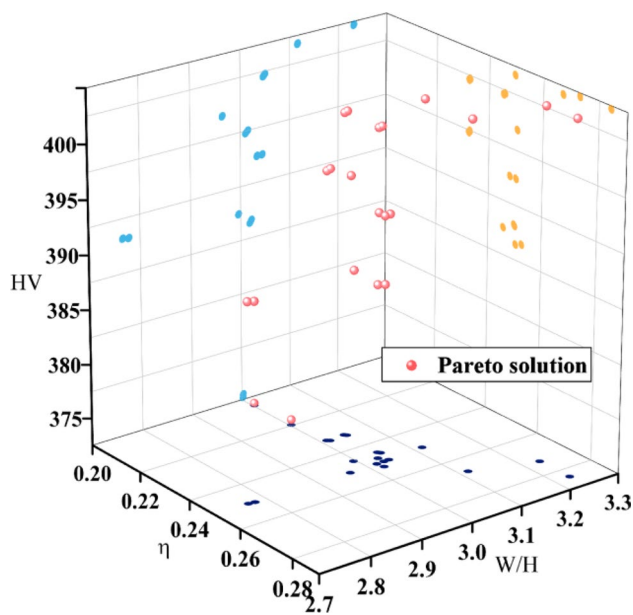


Fig. 12 Pareto front found by GNSGA-II for the laser cladding process

Table 7 Comparison of laser cladding performance before and after optimization

Indicators	η	W/H	HV
GNSGA-II	0.24525	3.04718	405
Experience value	0.16948	2.23630	359
Relative error	44.71%	36.26%	12.71%

(dilution rate, width-to-height ratio, average microhardness) was constructed by the GA-ACO-RFR algorithm, and the coating quality characteristics were optimized by the GNSGA multi-objective algorithm.

- (1) The regression prediction model based on GA-ACO-RFR has a good performance in predicting the coating quality characteristics, the validation results show that the error between the real values of the predicted values is no more than 5%.
- (2) The Pareto solution set obtained using the GNSGA-II multi-objective optimization algorithm with the introduction of the Gaussian distribution operator provides more flexibility and convenience in the selection of process parameters for machining practice.
- (3) The optimized laser cladding process parameters based on the Pareto solution set showed improved coating dilution, width-to-height ratio, and average microhardness compared with coating obtained from engineering experience.

With the development of intelligent algorithms, more and more algorithms with better performance are available, we will continue to explore more excellent intelligent algorithms for laser cladding technology. Different working conditions have different requirements for the surface modification of the parts and for the quality characteristics of the laser clad coatings, which will be explored for more coating quality characteristics in future studies. Meanwhile, multi-pass laser cladding coatings are mainly influenced by the process parameters and lap rates of single-track coatings. To obtain a good flatness of the multichannel coating, the effect of the lapping rate on the flatness of the coating will be subsequently investigated.

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Declarations

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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